

Technical response on c_δ and the robustified c_δ^*

Definitions, Pearson equivalence, Mantel comparison, and outlier status

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Main conclusion. The robust-centre statistic is best treated as a separate c_δ^* measuring concordance of observation-level salience profiles. It is not a robust substitute for every purpose of the original c_δ . For a fixed pair of profiles, c_δ^* and Pearson correlation give identical one-sided permutation evidence. Mantel instead asks whether the complete dyadic distance patterns agree. The three procedures therefore require different scientific wording.

1 Definitions and the change in research question

For observations $x_1, \dots, x_n$ and $y_1, \dots, y_n$, define the original one-dimensional L2 divergence profiles

$$d_{Xi} = \left\{ \frac{1}{n-1} \sum_{j \neq i} (x_i - x_j)^2 \right\}^{1/2}, \quad d_{Yi} = \left\{ \frac{1}{n-1} \sum_{j \neq i} (y_i - y_j)^2 \right\}^{1/2}. \quad (1)$$

The corrected coefficient is

$$c_\delta = \frac{n^{-1} \sum_i d_{Xi} d_{Yi}}{(n^{-1} \sum_i d_{Xi})(n^{-1} \sum_i d_{Yi})}. \quad (2)$$

It asks whether the same labelled observations have high overall divergence from their respective groups.

For the new statistic, let T_X, T_Y be Huber locations fitted using fixed MAD scales s_X, s_Y , and set

$$r_{Xi} = \frac{|x_i - T_X|}{s_X}, \quad r_{Yi} = \frac{|y_i - T_Y|}{s_Y}, \quad c_\delta^* = \frac{n^{-1} \sum_i r_{Xi} r_{Yi}}{\bar{r}_X \bar{r}_Y}. \quad (3)$$

This asks whether the same observations are salient or atypical in the two measurements. I therefore agree that the robust-centre version changes the research question and merits the notation c_δ^* . “Reproducible anomaly” or “salience-profile concordance” is a more accurate description than “robustified original c_δ ”.

There is still one important qualification. The Huber centre protects the reference point, but the uncapped radius scores every remote observation at its full distance. Thus c_δ^* is robust in centre fitting, not bounded in its final observation leverage.

2 Question 1 and 2: relationship with Pearson correlation

For any positive paired profiles a_i, b_i with nonzero variance,

$$\frac{\overline{ab}}{\bar{a}\bar{b}} = 1 + \text{Cor}(a, b) \text{CV}(a) \text{CV}(b). \quad (4)$$

Consequently,

$$\boxed{c_\delta^* = 1 + r_{\text{Pearson}}(r_X, r_Y) \text{CV}(r_X) \text{CV}(r_Y)}. \quad (5)$$

Under permutations of the pairing, the two CVs are fixed and positive. Therefore c_δ^* and Pearson correlation have exactly the same permutation ordering, one-sided permutation p -value, and rejection decision. They are not numerically identical estimands: c_δ^* combines concordance with the two marginal profile CVs, whereas Pearson isolates standardized concordance.

Multiplying either profile by a positive constant cancels from c_δ^* and leaves Pearson correlation unchanged. Hence using absolute deviations from the robust centre without MAD normalization gives exactly the same result.

Table 1: Numerical verification with 4,999 common permutations per scenario.

Scenario	c_δ^*	Pearson r	$CV_X CV_Y$	$p_{c_\delta^*} = p_r$
Normal correlated	1.1641	0.2805	0.5849	0.0068
t_3 correlated	1.5217	0.4571	1.1413	0.0012
Skew correlated	1.4197	0.3773	1.1124	0.0064
Matched outlier	3.4507	0.8893	2.7557	0.0014
Unmatched outlier	1.3508	0.1377	2.5474	0.0280

Across all five cases, the identity error, scale-normalization difference, permutation p -value difference, and maximum permutation-rank disagreement were zero up to floating-point precision.

Interpretive consequence. If the inferential question is only “are the two robust residual profiles positively concordant?”, Pearson plus the same permutation scheme is sufficient and more familiar. c_δ^* adds a distinct descriptive scale only if the product of marginal salience CVs is scientifically meaningful and should amplify or attenuate the concordance.

3 Original c_δ versus Mantel (1967)

Mantel’s procedure compares the corresponding entries of two complete dissimilarity matrices. In its common standardized form it correlates $\{D_{ij}^X : i < j\}$ with $\{D_{ij}^Y : i < j\}$, and obtains a reference distribution by simultaneously permuting rows and columns (unit labels) of one matrix [1, 2].

The original c_δ first compresses each distance matrix to one row summary per observation. In one-dimensional L2 form there is an exact identity

$$d_{Xi}^2 = \frac{n}{n-1} \left\{ (x_i - \bar{x})^2 + s_{X,\text{pop}}^2 \right\}. \quad (6)$$

Thus the entire row profile is a monotone function of absolute centred magnitude. It does not remember which other observations are above or below x_i , or which dyads generated the row total.

To demonstrate the distinction, 40 observations were arranged as 20 opposite-sign pairs with fixed absolute radii. Signs were independently rewired between X and Y while preserving each observation’s absolute centred radius. Equation (6) then makes the two L2 profiles identical for every replication.

Therefore Mantel does not make c_δ redundant, but c_δ cannot be described as preserving all pairwise differences. A precise statement is:

c_δ compares which labelled units have similar levels of overall divergence within their

Table 2: Information target of the two procedures.

	Preserves	Discards or does not isolate
Original c_δ	Labelled observation-level global divergence; whether the same rooms/persons are overall unusual	Exact dyadic geometry; which pairs are far apart; direction/sign in one dimension
Mantel/QAP	Correspondence of every labelled pairwise distance; dyadic distance geometry	A dedicated observation-level salience summary; ordinary Mantel does not solve spatial/block exchangeability automatically

Table 3: Same divergence profile, different dyadic geometry; 2,000 replications per row.

Pair-flip probability	Mean c_δ	Profile r	Mean Mantel r	Mantel SD
0.00	1.0418	1.0000	1.0000	0.0000
0.25	1.0418	1.0000	0.4297	0.1881
0.50	1.0418	1.0000	0.2308	0.0776
0.75	1.0418	1.0000	0.4170	0.1811
1.00	1.0418	1.0000	1.0000	0.0000

groups. Mantel compares whether the complete pattern of labelled pairwise distances agrees.

3.1 Across buildings

The claim that c_δ compares across buildings is defensible only if units have meaningful correspondence and the scientific target is node-level global divergence. It should not be stated as comparing the complete disease-versus-ventilation geometry across apartments.

Building heterogeneity must also be reflected in the permutation design. In 12 high-replication conditional-null conditions (1,500 datasets each), pooling and permuting across blocks produced severe false rejection, whereas within-building permutations restored calibration:

Table 4: Mean null rejection across heterogeneous-block conditions.

Method	Unrestricted permutation	Within-building permutation
Original L2 c_δ	0.4417	0.0481
Huber c_δ^*	0.5158	0.0474
Cap-6 sensitivity	0.5201	0.0473

Two separately calculated Mantel correlations should not simply be compared by an ordinary Fisher- z difference because dyadic entries are dependent. If the aim is to compare relationship strength across buildings, a unit/block-level permutation or bootstrap interaction test is preferable. If the aim is to explain a disease-distance matrix by ventilation distance plus building and other distance predictors, MRQAP is the closer framework [3]. Mantel and partial Mantel also require caution under spatial autocorrelation [4].

4 Status of meaningful outliers

I agree with the proposed distinction:

RQ1: exact group structure. If the exceptional member genuinely belongs to this particular team, its influence is part of the phenomenon. The original c_δ is a legitimate descriptive statistic, and strong L2 weighting may be intentional.

RQ2: stable or transferable structure. If the target is the ordinary or reproducible pattern beyond this exact group, leverage control and sensitivity analyses become necessary.

However, “the outlier belongs” and “its squared leverage is optimal” are different propositions. Inclusion is a scientific decision; the exponent or cap is an estimand-weighting decision.

The following experiment used 1,000 independent $n = 60$ datasets per condition. One high observation was inserted at the same index (matched) or different indices (unmatched). “Top share” is the largest single contribution to the numerator divided by the entire numerator.

Table 5: Effect of a meaningful extreme observation.

Condition	Method	Magnitude	Mean coefficient	Profile r	Median top share
Matched	Original L2	8	1.200	0.852	0.273
Matched	Huber uncapped c_δ^*	8	2.028	0.709	0.632
Matched	Huber cap 6	8	1.580	0.560	0.496
Matched	Original L2	32	1.530	0.997	0.474
Matched	Huber uncapped c_δ^*	32	10.295	0.979	0.965
Matched	Huber cap 6	32	1.582	0.560	0.497
Unmatched	Original L2	32	0.991	-0.018	0.101
Unmatched	Huber uncapped c_δ^*	32	0.825	-0.018	0.377
Unmatched	Huber cap 6	32	0.984	-0.015	0.139

The original L2 outlier changes ordinary members’ divergence scores because every row contains a distance to it. At magnitude 32, the mean relative shift among ordinary L2 profile entries was about 2.28. For Huber radii it was only about 0.046: the centre is protected and spillover is small. Yet the uncapped matched observation itself then dominates the product numerator. This separates two kinds of robustness:

1. **reference robustness:** ordinary observations are not moved much;
2. **final leverage robustness:** no single observation dominates the coefficient.

Uncapped c_δ^* has the first property but not the second. Cap 6 is a separate, bounded sensitivity estimand that provides the second property.

5 Recent inferential findings

For the uncapped Huber c_δ^* , a complete functional-delta influence function has been derived. It includes the median and MAD effects on both Huber centres, the numerator and two denominator moments, and all covariance induced by the paired (X_i, Y_i) observations. Under iid pairs, continuous positive marginal densities at the median/MAD points, unique positive MADs and Huber roots, no mass at derivative knots, positive radius means, and finite joint squared radius-product moments,

$$\sqrt{n}(\hat{c}_\delta^* - c_\delta^*) = \frac{1}{\sqrt{n}} \sum_{i=1}^n IF_{c_\delta^*}(X_i, Y_i) + o_P(1) \implies N(0, V). \quad (7)$$

Population-level skew-lognormal quadrature matched regular-point contamination derivatives to scaled error below 1.2×10^{-5} . The theory itself is supported. Finite-sample studentization is not yet reliable: true analytic densities and cross-fitted KDE changed mean SE by at most 0.2%; HC-style correction remained below nominal coverage; and complete-refit bootstrap- t did not improve coverage. Therefore permutation inference remains the formal default, with the sandwich interval labelled exploratory.

6 Recommended terminology and reporting

1. Keep **original** c_δ : exact-set L2 divergence-profile concordance. State explicitly that it summarizes row-level/global divergence, not full dyadic geometry.
2. Introduce c_δ^* : Huber-reference absolute-salience-profile concordance. Avoid claiming bounded robustness for the uncapped version.
3. Report **Pearson robust-profile correlation** alongside c_δ^* , or use Pearson as the inferential statistic, because the permutation test is identical. Retain c_δ^* only when its $CV_X CV_Y$ weighting is substantively desired.
4. Keep **cap-6** c_δ^* as a pre-specified sensitivity estimand, not a post-hoc correction and not the same research question as the original c_δ .
5. Use **Mantel/QAP or MRQAP** when the hypothesis concerns complete dyadic distance correspondence or multiple distance-matrix predictors.
6. For multiple buildings, define whether the target is pooled, within-building, or an interaction before choosing the permutation scheme.

7 Immediate next checks

- Decide whether the $CV_X CV_Y$ weighting in Equation (4) is part of the intended construct. This is now the most important definition decision for c_δ^* .
- Test Mantel, profile Pearson, original c_δ , and c_δ^* on one shared building-style data generator with dyadic versus node-salience alternatives.

- Formalize an across-building interaction/permutation design using matched rooms or explicitly unmatched units.
- Derive the cap-6 influence function separately if bounded final leverage is to be claimed formally.

References

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